EcoTrack Dashboard

Product Requirements Document — Frontend UI Contest Submission

A gamified personal carbon-footprint tracker · React + Tailwind CSS · GSAP ScrollTrigger + Lenis

Field	Value
Author	Vishal (vishal.cj.cbe@gmail.com)
Document status	Draft v1.0 — ready for build
Target	Frontend-only UI/UX contest submission (no backend required for judging; mock data acceptable)
Primary stack	React 18 + Tailwind CSS · Framer Motion (micro-interactions) · GSAP + ScrollTrigger + Lenis (scroll-scrub)

1. Overview & Problem Statement

EcoTrack Dashboard is a personal carbon-footprint tracker that turns daily eco-friendly habits — recycling, using public transit, saving energy — into a game. Instead of a dry logging tool, it should feel like a living, breathing eco-companion: users log actions, watch a visual “impact score” grow, unlock streaks and badges, and see their footprint trend down over time through clear, satisfying charts.

Problem: Most carbon trackers are spreadsheet-like and boring, so people abandon them within days. There is no immediate emotional reward for a small eco-action, and data is usually shown as raw numbers rather than a story.

Opportunity (contest angle): This is a UI/UX contest — the winning factor is visual polish, motion craft, and a scroll experience that makes judges stop scrolling to watch. The PRD below is written to maximize that: a strong theme identity, purposeful micro-interactions, and one signature scroll-scrubbed hero sequence that demonstrates technical craft.

2. Goals & Success Metrics

- **Delight on first load:** hero section with a scroll-scrub animation that communicates the product idea in under 8 seconds of scrolling.
- **Gamification feels earned:** every logged action gives instant, animated feedback (points, streak, badge progress).
- **Data clarity:** the dashboard answers “am I improving?” in one glance via a hero chart + 3 supporting visualizations.
- **Contest-ready polish:** consistent theme tokens, accessible contrast, smooth 60fps animation, responsive across desktop/tablet/mobile.
- **Judging metrics:** visual cohesion, animation quality/purposefulness, originality of the scroll-scrub concept, usability of the habit-logging flow, and code/design-system tidiness.

3. Target Users / Personas

Persona	Motivation	Needs from the UI
Maya, 27 — Eco-conscious urbanite	Wants to see her transit/recycling habits add up to something measurable	Fast daily logging, visible streaks, satisfying micro-rewards
Dev, 34 — Competitive gamifier	Motivated by leaderboards, badges, levels	Clear progression system, badge gallery, level-up animation
Asha, 41 — Data-curious parent	Wants to understand household impact trends over months	Clean charts, comparisons (this week vs last), category breakdown
Contest judge	Evaluates craft, originality, usability in ~2 minutes	Strong first-scroll impression, obvious visual hierarchy, no clutter

4. Information Architecture (Screens)

- **Landing / Story Scroll** — single scroll-scrubbed hero sequence introducing the concept (Earth → habits → impact score).
- **Dashboard (Home)** — hero impact score ring, weekly trend chart, category breakdown, streak + badge strip, quick-log actions.
- **Log Activity** — modal / drawer to log recycling, transit, or energy-saving actions with instant point animation.
- **Progress & Insights** — monthly trend line, category comparison, CO■ saved equivalents (trees planted, km not driven).
- **Achievements** — badge gallery, streak calendar, level progression bar.
- **Profile / Settings** — theme toggle (light/dark), units, notification preferences.

5. Feature Requirements

5.1 Habit Logging & Gamification Core

- Three core habit categories: Recycling ■■, Public Transit ■■, Energy Saving ■ — each with 3–5 quick-tap presets (e.g. “Bus commute”, “Recycled plastics”, “Turned off standby devices”).
- Each logged action awards **Eco Points** with an animated count-up + a floating “+XX pts” particle burst.
- **Daily streak counter** with a flame/leaf icon that grows visually the longer the streak continues.
- **Level system** (Sprout → Sapling → Tree → Forest Guardian) with a progress bar that animates toward the next threshold.
- **Badge system**: unlockable achievement cards with a flip/reveal animation on unlock (e.g. “7-Day Streak”, “Zero-Car Week”).
- Optional social proof: a lightweight, local (mocked) leaderboard — no auth required for the contest build.

5.2 Carbon Footprint Tracking

- Convert logged actions into estimated CO■ saved (kg), using simple static multipliers per action type (documented in a constants file, not a live API — fine for a frontend contest).
- Show cumulative “Impact Score” (points) and “CO■ Saved” (kg) as the two hero numbers on the dashboard.
- Translate CO■ saved into relatable equivalents (e.g. “= 3 trees planted”, “= 42 km not driven”) shown as animated icon chips.

6. Data Visualization Requirements

Chart	Purpose	Type / Library	Animation
Impact Score Ring	Hero KPI: today's/weekly progress toward goal	Radial progress ring (SVG + Framer Motion)	Ring sweeps from 0 to value on load; glow pulse on milestone
Weekly Trend	Is footprint reduction trending down?	Area/line chart (Recharts or Visx)	Line draws left-to-right (stroke-dashoffset), area fades in
Category Breakdown	Which habit contributes most impact	Donut chart, 3 segments (Recycling/Transit/Energy)	Segments sweep in sequence, hover expands segment
Streak Calendar	Consistency at a glance (GitHub-style heatmap)	7x?? grid of cells, color intensity = activity	Cells stagger-fade in on mount, ripple on hover
CO■ Equivalents	Make abstract kg CO■ feel tangible	Icon-stat chips (tree, car, lightbulb)	Count-up numbers + icon bounce-in

All charts use the theme's semantic color tokens (below) so the palette stays consistent even as the library renders SVG paths.

7. UI Theme System — “Organic Eco”

Primary direction: warm, natural, rounded — forest greens and moss tones on a soft cream base, with a sky-blue secondary accent for data/links. This reads instantly as “eco” without leaning on cliché neon-green tech aesthetics. Two alternate directions are listed at the end of this section if a different contest tone is wanted.

7.1 Color Tokens

Token	Hex	Usage
--color-forest (primary)	#1F4D36	Headers, nav, primary CTA background
--color-moss (primary-alt)	#4A7C59	Secondary buttons, active states, icons
--color-leaf (accent)	#8FBF7F	Progress fills, success states, badges
--color-sky (data accent)	#5E9EA3	Charts' secondary series, links, info states
--color-cream (surface)	#F7F3E9	App background (light mode)
--color-card (surface-2)	#FFFFFF	Card backgrounds, elevated surfaces
--color-ink (text)	#2B2B26	Primary text
--color-muted (text-2)	#6B6B5F	Secondary text, captions
--color-warn (alert)	#D98E4A	High-footprint warnings, gentle nudges
Dark mode surfaces	#10241A / #16311F	App bg / card bg when dark mode toggled

7.2 Typography

- **Display / headings:** “Clash Display” or “Sora” (geometric, friendly, rounded terminals) — weight 600–700.
- **Body:** “Inter” or “General Sans” 400/500 — excellent legibility at small sizes for stat labels.
- **Numerals for stats:** tabular-nums so count-up animations don't jitter in width.
- Type scale: 12 / 14 / 16 / 20 / 28 / 40px, 1.25 line-height for body, 1.1 for display.

7.3 Shape & Surface Language

- Corner radius scale: 8px (chips) / 16px (cards) / 24px (hero panels) — consistently rounded, never sharp.
- Cards: soft 1-2 layer shadow (0 4px 16px rgba(31,77,54,0.08)), subtle 1px moss-tinted border on hover.
- Organic accents: blurred leaf-shaped blob shapes (SVG) as background decoration behind hero stats — low opacity, parallax on scroll.
- Iconography: rounded line icons (Phosphor Icons or Lucide, “duotone” style) tinted with the leaf/moss palette.

7.4 Alternate Theme Directions (backup options)

Theme	Feel	When to use instead
Glass / Aurora	Dark base, frosted glass cards, green-teal gradient glow	If you want a more “tech dashboard”, high-drama contest look
Neo-Brutalist Eco	Bold black borders, hard offset shadows, lime accents	If you want maximum memorability / stands out visually

Theme	Feel	When to use instead
Data-Clean Minimal	Neutral gray base, single green accent, lots of whitespace	If judging criteria reward restraint & information clarity

8. Button Themes

Variant	Style	State behavior
Primary	Filled forest-green bg, white text, 16px radius, subtle drop shadow	Hover: lightens to moss + lifts 2px (translateY) over 150ms ease-out. Active: presses down, shadow flattens.
Secondary	Outline moss border, transparent bg, moss text	Hover: fills with 10% moss tint. Focus: 2px sky focus ring for accessibility.
Ghost / Text	No border, moss text only — used in nav/toolbar	Hover: underline slides in left-to-right (150ms), text darkens to forest.
Action / Log (FAB)	Circular, gradient leaf→moss, icon-only, elevated	Hover: scale 1.06 + shadow bloom. Tap: quick squash-stretch (0.92→1 scale, spring).
Success / Confirm	Filled leaf-green, used after logging an action	On click: ripple burst + brief confetti-particle emit (CSS/canvas), then morphs into a checkmark icon.
Disabled	10% opacity, no shadow, cursor not-allowed	No hover/active transitions.

All interactive states use a shared 150–200ms cubic-bezier(0.4, 0, 0.2, 1) easing token (`--ease-standard`) so the whole app feels like one consistent material.

9. Animation System

9.1 Principles

- **Purposeful, not decorative:** every animation should confirm an action, guide attention, or celebrate progress — never just “move for movement's sake.”
- **Consistent timing:** micro (100–200ms) for hover/tap, standard (300–500ms) for panel/card transitions, hero (800–2000ms, scroll-driven) for the story sequence.
- **Reduced-motion respect:** honor `prefers-reduced-motion` — swap scrub/parallax for simple fades.

9.2 Micro-interactions (Framer Motion)

Element	Trigger	Animation
Nav links	Hover	Underline grows from center, color shifts to forest
Stat cards	Mount	Stagger fade + slide-up (y: 12→0), 60ms stagger between cards
Impact Score ring	Data update	Ring animates to new value via animated stroke-dashoffset, spring easing
Habit log button	Tap	Squash-stretch + point burst particles + hapt-style ripple
Badge unlock	New badge earned	Card flips 180° (rotateY), glow pulse, confetti burst, toast slides in from top
Streak flame/leaf	Streak increments	Icon scales up briefly (1→1.3→1) with a warm glow flash
Toast notifications	Any success event	Slide + fade in from top-right, auto-dismiss slide-out after 3s

9.3 Page & Panel Transitions

- Route/tab changes: crossfade + 8px slide (200ms) via Framer Motion's AnimatePresence.
- Modal/drawer (Log Activity): slide up from bottom on mobile, scale+fade from center on desktop, with a backdrop blur fade-in.

10. Scroll-Scrub Animation Spec (Signature Feature)

This is the centerpiece for contest impact: a scroll-scrubbed hero story on the landing page, where scroll position directly drives an animation timeline (not just triggers it) — the user “scrubs” through the narrative by scrolling, like a video timeline bound to the scrollbar.

10.1 Tech Approach

- **Lenis** for buttery smooth/inertial scrolling (wraps native scroll, feeds normalized scroll value).
- **GSAP ScrollTrigger** with `scrub: true` (or a numeric value like 0.6 for slight lag/smoothing) binding a GSAP timeline's progress directly to scroll position within a pinned section.
- Section is `pin: true` for its scroll duration (e.g. 300vh of scroll = one pinned viewport playing the full sequence), then releases into normal scroll for the rest of the landing page.

10.2 Narrative Sequence (single pinned hero, ~3x viewport height of scroll distance)

Scroll %	Visual beat
0–15%	Earth illustration + headline fade/scale in: “Your habits, visualized.”
15–40%	Three habit icons (recycle/transit/energy) fly in from edges and orbit the Earth
40–65%	Icons morph into data points; a line chart draws itself, footprint line trending down
65–85%	Impact Score ring sweeps to 87%, count-up numbers animate, particles emit
85–100%	Content settles into the actual dashboard preview card + CTA button fades/slides in, unpins into normal scroll

10.3 Supporting Scroll Effects (rest of page, non-pinned)

- Parallax leaf/blob background shapes move at 0.3–0.6x scroll speed for depth.
- Feature cards further down use standard ScrollTrigger (not scrubbed) with `toggleActions` for a simple stagger-reveal on enter.
- A slim progress bar at the top of the viewport fills as the user scrubs through the hero sequence, giving a clear sense of “how much story is left.”

10.4 Performance Guardrails

- Animate only `transform` and `opacity` inside the scrubbed timeline (GPU-accelerated, no layout thrash).
- Use `will-change: transform` sparingly on the pinned section's direct children only.
- Provide a static, single-viewport fallback (fade-in only) when `prefers-reduced-motion: reduce` is set.

11. Accessibility & Responsive Requirements

- Color contrast: all text/background pairs meet WCAG AA (4.5:1 body, 3:1 large text) — verify forest-on-cream and white-on-forest combos.
- All interactive elements reachable and operable via keyboard; visible focus ring using the sky accent (2px, 2px offset).
- Charts include a text/table alternative (visually hidden or expandable) summarizing the same data for screen readers.
- Scroll-scrub hero degrades to simple fade-in sequence under `prefers-reduced-motion` and remains fully readable without JS animation.
- Responsive breakpoints: mobile (<640px, single column, FAB for quick-log), tablet (640–1024px, 2-col dashboard), desktop (>1024px, full 3-col dashboard with sidebar nav).

12. Tech Stack & Libraries

Layer	Choice	Notes
Framework	React 18 (Vite)	Fast dev server, ideal for a contest build/demo
Styling	Tailwind CSS + CSS variables for theme tokens	Utility classes + design tokens from Section 7
Micro-interactions	Framer Motion	AnimatePresence for transitions, layout animations for cards
Scroll-scrub	GSAP + ScrollTrigger + Lenis	Pinned timeline per Section 10
Charts	Recharts (or Visx for custom SVG control)	Wrap with theme tokens; animate via GSAP where Recharts' built-ins are too limited
Icons	Lucide or Phosphor Icons	Rounded, duotone-tintable
State	Zustand or React Context	Local-only state; mock data / localStorage for contest demo, no backend required
Fonts	Sora / Clash Display (display) + Inter (body)	Loaded via Google Fonts / Fontshare

13. Acceptance Criteria & Judging Fit

- ✓ Landing page opens with the scroll-scrub hero sequence; scrolling up/down scrubs it forward/backward smoothly.
- ✓ Dashboard loads with animated count-up stats and at least the 3 core charts from Section 6.
- ✓ Logging any habit action produces the full feedback loop: point burst → count-up → streak update → (if applicable) badge unlock animation.
- ✓ Theme tokens are centralized (Tailwind config / CSS variables) — no hardcoded hex values scattered in components.
- ✓ All animations run at 60fps on a mid-range laptop; reduced-motion fallback verified.
- ✓ Fully responsive from 375px to 1440px+ widths.
- ✓ Deployed to a public URL (Vercel/Netlify) for judges to interact with live — not just screenshots.

End of document — EcoTrack Dashboard PRD v1.0